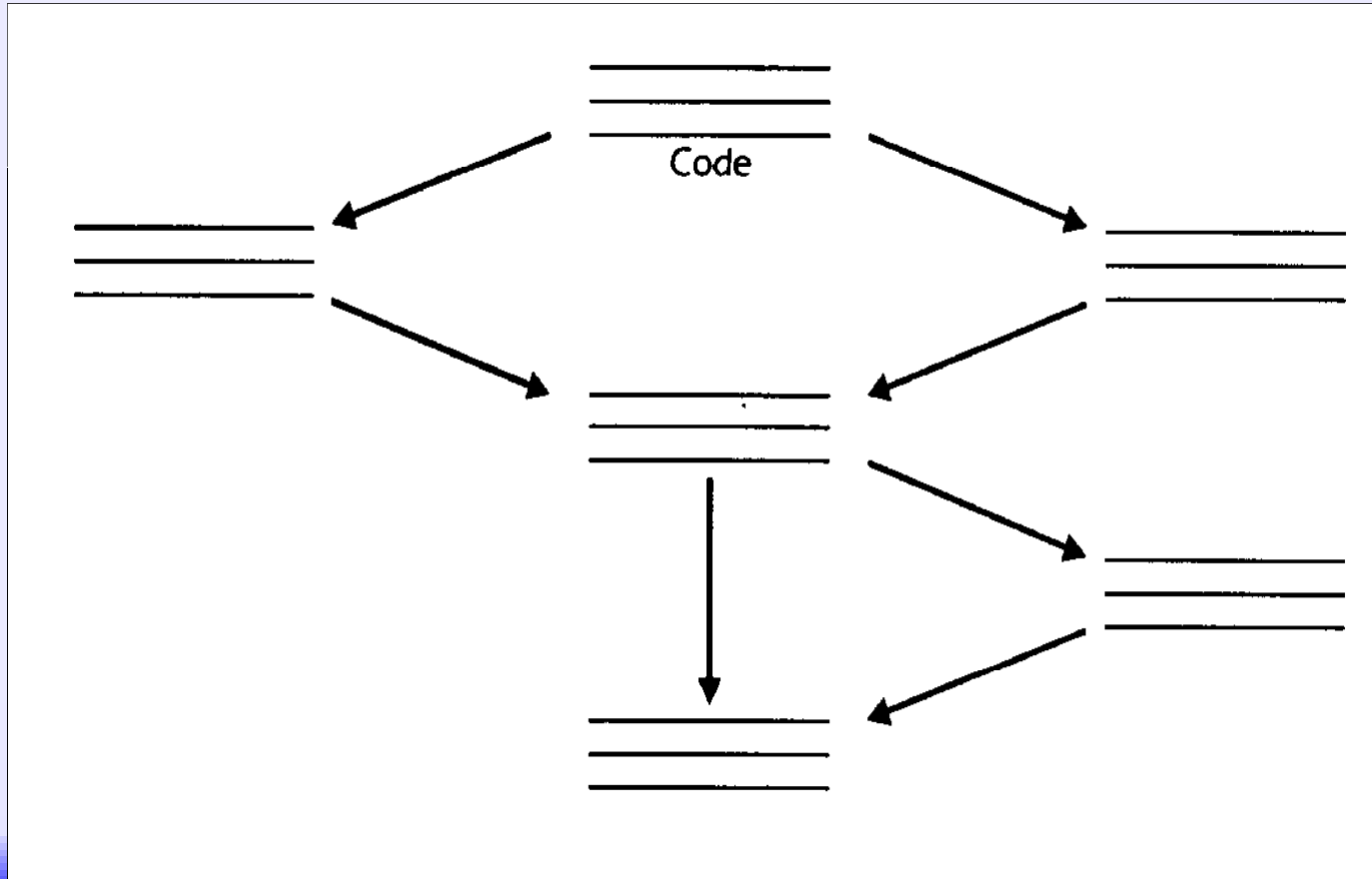


C023115
Software Testing- Unit 2
(12M)



White box testing

- Look inside a component and create tests based on implementation



White Box Testing

- White box testing is a testing technique, that examines the program structure and derives test data from the program logic/code
- White box testing is a strategy in which testing is based on:
 - the internal paths,
 - structure, and
 - implementation of the software under test (SUT).
- Also known as ***Structural testing, Clear box Testing, and Glass box Testing***
- Requires detailed programming skills



White Box Testing

Test Cases are designed to:

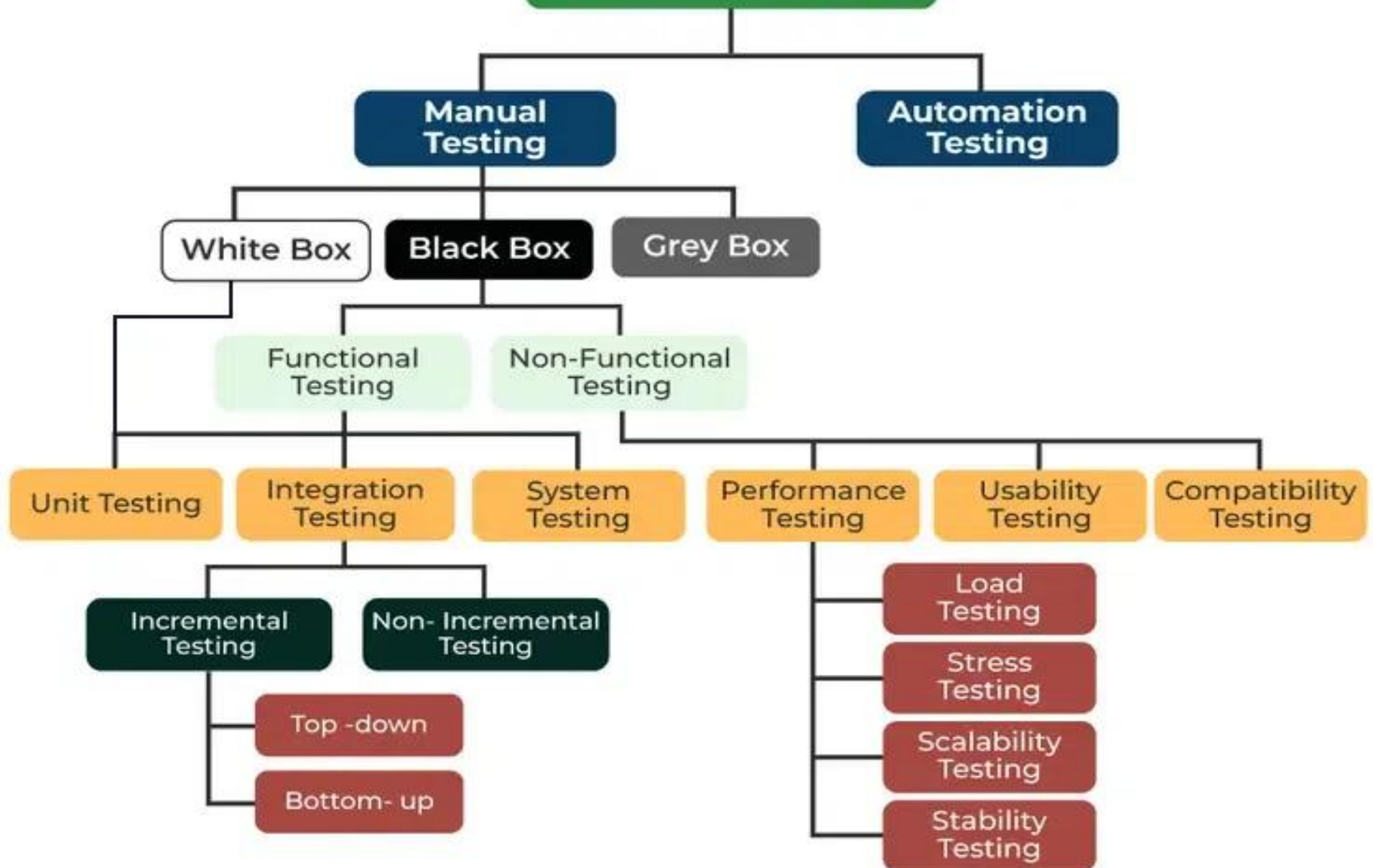
- (1) exercise **Independent paths** within a module or unit;
- (2) exercise **logical decisions** on both their true and false side;
- (3) execute **loops** at their boundaries and within their operational bounds; and
- (4) exercise **internal data structures** to ensure their validity



Why white box techniques?

- Find bugs early
- Find bugs that are hard to find with BB techniques
- Provides good ideas to BB testers

Types of Software Testing



Advantages of White box Testing

Thorough Testing: White box testing is thorough as the entire code and structures are tested.

Code Optimization: It results in the optimization of code removing errors and helps in removing extra lines of code.

Early Detection of Defects: It can start at an earlier stage as it doesn't require any interface as in the case of black box testing.

Integration with SDLC: White box testing can be easily started in the [Software Development Life Cycle](#).

Detection of Complex Defects: Testers can identify defects that cannot be detected through other testing techniques.

1. Path Testing

- White box testing will be checks all possible execution paths in the program to sure about the each one of the function behaves as expected.
- It helps verify that all logical conditions in the code are functioning correctly and efficiently with as properly manner, avoiding unnecessary steps with better code reusability.

Advantages of Path Testing

- The path testing method reduces the redundant tests.
- Path testing focuses on the logic of the programs.
- Path testing is used in test case design.



2. Loop testing

- Loop Testing is a type of software testing that is performed to validate the loops. It is one of the type of Control Structure Testing. Loop testing is a white box testing technique and is used to test loops in the program.
- It will be check that loops (for or while loops) in the program operate correctly and efficiently. It checks that the loop handles variables correctly and doesn't cause errors like infinite loops or logic flaws.

Advantages of Loop Testing

- Loop testing limits the number of iterations of loop.
- Loop testing ensures that program doesn't go into infinite loop process.
- Loop testing endures initialization of every used variable inside the loop.
- Loop testing helps in identification of different problems inside the loop.

Loop testing helps in determination of capacity.

Unit Testing as White-box Testing

- Unit Testing focuses on testing individual units or components of the software (such as functions or methods) by directly checking the code's internal logic, flow, and behavior.



4. Integration Testing as White-box Testing

- Integration Testing can be White-box Testing when it focuses on verifying the internal logic and interactions between components or modules.
- In this case, the tester needs to understand the internal code and logic of the different modules being integrated



White Box Testing Focus On?

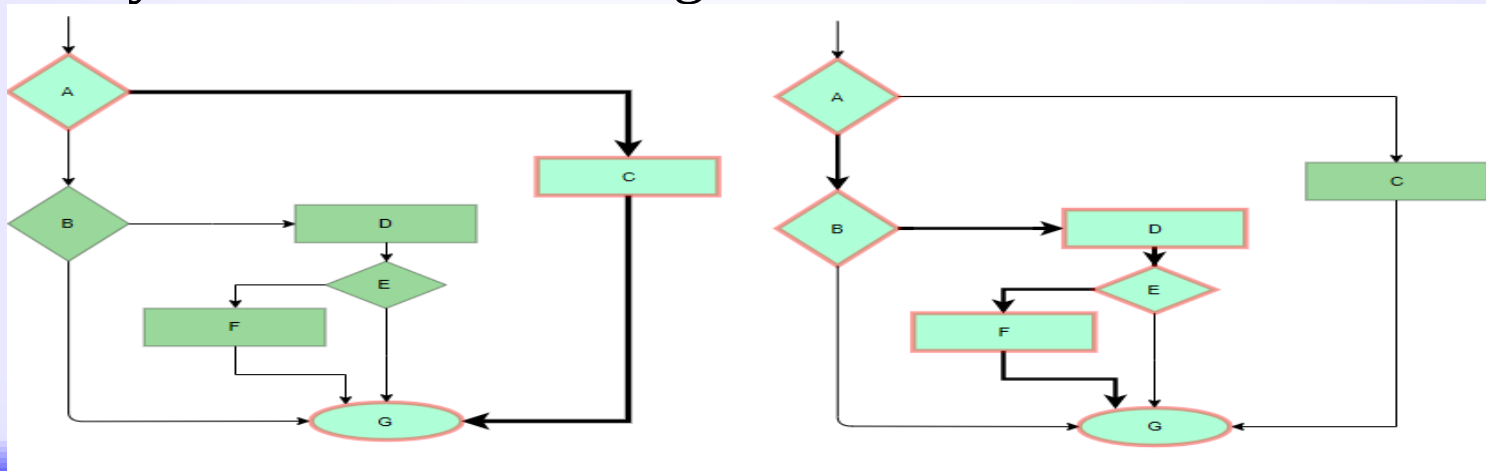
- White-box Testing focuses on the internal workings of an application, ensuring that its logic, structure, and flow operate as intended.
- Unlike black-box testing, which focuses on user interactions without knowledge of the underlying code, white-box testing involves examining the software's source code directly.

White Box Testing Techniques



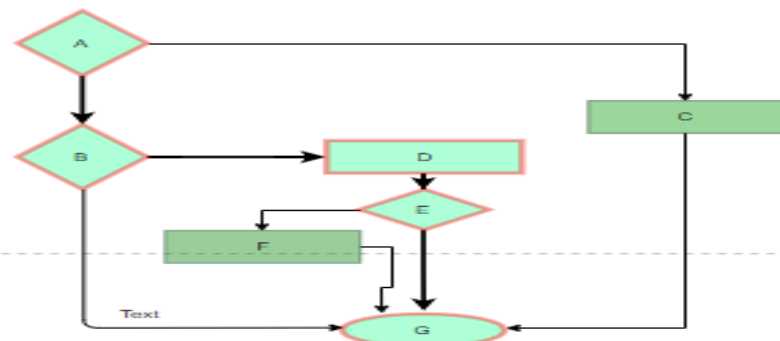
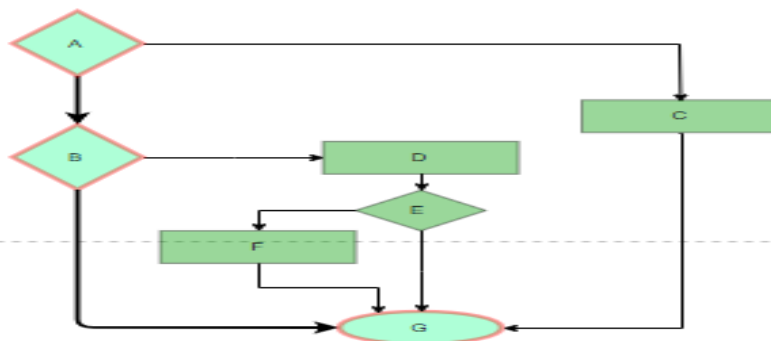
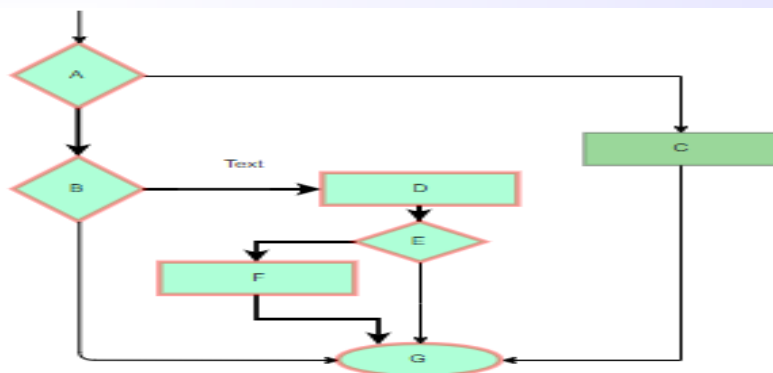
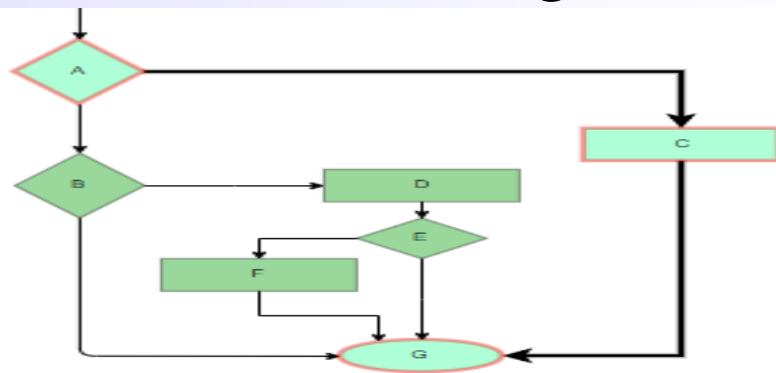
1. Statement Coverage

- In this technique, the aim is to traverse all statements at least once.
- Hence, each line of code is tested. In the case of a flowchart, every node must be traversed at least once.
- Since all lines of code are covered, it helps in pointing out faulty code.
- If we see in the case of a flowchart, every node must be traversed at least once.
- Since all lines of code are covered, it helps in pointing out faulty code while detecting.



2. Branch Coverage

- Branch coverage focuses on testing the decision points or conditional branches in the code.
- It checks whether both possible outcomes (true and false) of each conditional statement are tested.
- In this technique, test cases are designed so that each branch from all decision points is traversed at least once.
- In a flowchart, all edges must be traversed at least once.



3. Decision Coverage

- The software decision coverage testing is a part of the white box testing. It is conducted by the developers to verify the total decision results in the program.
- It only deals with the boolean values, namely true or false to every boolean expression.
- In some scenarios wherever there are probabilities of two or many outcomes originating from the statements such as do while, if, and case, it is referred to as decision points as there are two outcomes either false or true.
- **Formula to Calculate the Software Decision Coverage Percentage**
- The software decision coverage can be calculated by dividing the total count of results executed with the total count of decision results in the source code, and then multiplied by hundred.
- **Decision Coverage = (Total count of decision results executed / Total count of decision results in the source code) * 100**

4. Control Flow Testing

- **It is** a type of software testing that uses a program's control flow as a model.
- Control flow testing is a structural testing strategy. This testing technique comes under white box testing.
- For the type of control flow testing, all the structure, design, code and implementation of the software should be known to the testing team.
- This type of testing method is often used by developers to test their own code and own implementation as the design, code and the implementation is better known to the developers. .



5 Code Logic and Flow

- Checks if the program's logic works as intended.
- This means verifying that code modules (like functions or classes) interact correctly and that control structures such as if-else statements, loops, or switches execute properly.
- For example, ensuring a login function redirects users correctly based on valid or invalid credentials.

6. Data Flow Testing

- Data Flow Testing is a structural testing method that examines how variables are defined and used throughout a program.
- It uses control flow graphs to identify paths where variables are defined and then utilized, aiming to uncover anomalies such as unused variables or incorrect definitions.
- By focusing on the flow of data, it helps ensure that variables are properly handled and used in the code.
- It is a method that is used to find the test paths of a program according to the locations of definitions and uses of variables in the program.
- By analyzing control flow graphs, this technique aims to identify issues such as unused variables or incorrect definitions, ensuring proper handling of data within the code.



6. Data Flow Testing

- **Advantages of Data Flow Testing:**
- Data Flow Testing is used to find the following issues-
- To find a variable that is used but never defined,
- To find a variable that is defined but never used,
- To find a variable that is defined multiple times before it is use,
- Deallocating a variable before it is used.
- **Disadvantages of Data Flow Testing**
- Time consuming and costly process
- Requires knowledge of programming languages

